

TLQLDMR: A Transfer Learning Quantile Regression Model for Wind Power Forecasting under Extreme Weather Conditions

***a, ***a, ***b, ***a and ***a,*

^aSchool of Artificial Intelligence, Southwest University, Chongqing 400715, China

^bBusiness school, Jiangsu open University, Jiangsu 210036, China

ARTICLE INFO

Keywords:

Wind power forecasting
Large margin distribution machine
Quantile Regression
Transfer learning

ABSTRACT

In the current context of new energy power systems, wind power generation forecasting under extreme weather conditions faces severe challenges such as historical sample scarcity and significant data distribution shifts. Traditional support vector regression (SVR) based prediction models are mostly based on the assumption that the training and testing data are evenly distributed, which makes it difficult to effectively deal with the distribution differences between the source domain (conventional weather) and the target domain (extreme weather). Therefore, in practical applications, the generalization performance is limited, and the prediction accuracy decreases significantly. To address the above issues, this paper proposes to extend the Large Margin Distribution Machine for Regression (LDMR) to transfer learning frameworks and construct a probability prediction model with cross domain adaptability. On the basis of inheriting LDMR to optimize the marginal distribution variance and mean, thereby enhancing the model's generalization ability, this model introduces a domain adaptation mechanism to explicitly reduce the feature distribution difference between the source domain and the target domain. In addition, by introducing quantile regression strategy, point prediction can be extended to probability prediction, which can quantify the uncertainty in wind power output in the form of prediction intervals, providing richer decision-making basis for power grid scheduling and risk prevention under extreme weather conditions.

1. Introduction

2. Related Work

3. Proposed Methodology

In this section, we elaborate on the construction of the Transfer Learning Quantile Regression Model (TLQLDMR). We first define the necessary notations, then present the final objective function which integrates Structural Risk, Distribution Variance, Domain Adaptation (MMD), and Weighted Pinball Loss. Finally, we describe the constraints introduced to handle the non-differentiability of the Quantile Loss.

3.1. Notations

In the field of machine learning, a domain $\mathcal{D}$ consists of a feature space $\mathcal{X}$ and a marginal probabilistic distribution $P(\mathbf{X})$ on the feature space, i.e., $\mathcal{D} = \{\mathcal{X}, P(\mathbf{X})\}$. Given a domain, a task $\mathcal{T}$ consists of a label space $\mathcal{Y}$ and an objective prediction function $f(\cdot)$, represented as $\mathcal{T} = \{\mathcal{Y}, f(\cdot)\}$. From the perspective of probability theory, the objective prediction function can be expressed as $P(\mathbf{Y}|\mathbf{X})$, and the task is represented as $\mathcal{T} = \{\mathcal{Y}, P(\mathbf{Y}|\mathbf{X})\}$.

In the problem of wind power forecasting under extreme weather, there are two crucial concepts corresponding to the different data distributions. The **Source Domain (Normal Weather)** is represented as $\mathcal{D}_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, containing a large amount of conventional weather samples. The **Target Domain (Extreme Weather)** is represented as $\mathcal{D}_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, containing a small number of extreme weather samples. Here, the input space $\mathcal{X} \subseteq \mathbb{R}^d$ represents the feature vectors constructed after sliding window processing, and the output space $\mathcal{Y} \subseteq \mathbb{R}$ represents the wind power. n_S and n_T denote the number of samples in the source and target domains, respectively, satisfying $n_T \ll n_S$.

*Corresponding author: ***

E-mail address: ***

ORCID(s):

Although the source and target domains belong to the same type of forecasting task, their data distributions exhibit distinct characteristics due to the drastic meteorological changes during extreme events, i.e., $P_S(\mathbf{X}_S, \mathbf{Y}_S) \neq P_T(\mathbf{X}_T, \mathbf{Y}_T)$. This distribution discrepancy implies that a model trained solely on P_S will fail on P_T . We define a feature mapping $\phi : \mathcal{X} \rightarrow \mathcal{H}$ that maps the input to a high-dimensional Reproducing Kernel Hilbert Space (RKHS). The prediction model is defined as:

$$f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b, \quad (1)$$

where $\mathbf{w}$ is the weight vector and b is the bias. The quantile $\tau \in (0, 1)$ is set to a specific value (e.g., $\tau = 0.95$) for interval prediction (e.g., predicting the upper bound).

3.2. TLQLDMR Model Formulation

3.2.1. Final Objective Function

Our optimization goal is to find the optimal $\mathbf{w}, b$ and slack variables ξ, ξ^* to minimize the following objective function J :

$$\begin{aligned} \min_{\mathbf{w}, b, \xi, \xi^*} J = & \underbrace{\frac{1}{2} \|\mathbf{w}\|^2}_{(1) \text{ Structural Risk}} \\ & + \lambda_1 \underbrace{\sum_{i=1}^{n_S} ((y_{S,i} - \mathbf{w}^T \phi(\mathbf{x}_{S,i}) - b) - \bar{u}_S)^2}_{(2) \text{ Distribution Variance (LDMR Core)}} \\ & + \lambda_2 \underbrace{\left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_{\mathcal{H}}^2}_{(3) \text{ Domain Adaptation (MMD)}} \\ & + \underbrace{C_S \sum_{i=1}^{n_S} L_{\tau}(y_{S,i}, f(\mathbf{x}_{S,i})) + C_T \sum_{j=1}^{n_T} L_{\tau}(y_{T,j}, f(\mathbf{x}_{T,j}))}_{(4) \text{ Weighted Pinball Loss}} \end{aligned} \quad (2)$$

The physical meaning of each term is as follows:

1. **Structural Risk:** This term controls the complexity of the model and prevents overfitting, embodying the maximum margin principle of SVM.
2. **Distribution Variance:** This term is the core of our proposed Quantile LDMR (QLDMR), incorporating a critical modification to the original LDMR variance.
 - **Theoretical Conflict:** The original LDMR minimizes the variance of absolute residuals, which assumes a symmetric error distribution suitable for mean regression. However, this symmetry contradicts the inherent asymmetry of Quantile Regression (governed by τ). Applying a symmetric variance penalty would force the residual distribution towards the center, degrading the skewness required for accurate quantile estimation.
 - **Asymmetric Modification:** To resolve this, we employ an **Asymmetric Variance** regularization based on the algebraic (signed) residuals. This design allows the model to reduce error dispersion (compactness) without constraining the distribution's direction, thereby preserving the non-symmetric properties essential for the Pinball Loss and improving interval coverage.
3. **Domain Adaptation (MMD):** In the context of wind power forecasting under extreme weather conditions, our primary objective is to mitigate the differences in data distributions across various domains by means of learning a feature transformation. Consequently, the effective measurement of distribution disparities between domains becomes a paramount concern. In this pursuit, it is crucial to select a method suitable for quantifying distribution dissimilarity, while ensuring that the chosen method is computationally efficient and easily integrated into the optimization problem of our model. Maximum Mean Discrepancy (MMD) has been demonstrated as an effective approach for estimating the distance between two distributions, offering a relatively straightforward implementation with fast computation speed. It targets the input space $\mathcal{X}$ to minimize the distance between the distribution centers of the source and target domains.

4. **Weighted Pinball Loss:** This term targets the output space $\mathcal{Y}$ and acts as the decisive component for “Supervised Transfer Learning”. Unlike unsupervised transfer which relies solely on feature alignment, we explicitly incorporate the target domain samples D_T into the loss calculation for the following reasons:

- **Corrective Anchors:** The target samples act as “anchors for correction”. The source domain data (D_S) teaches the model general laws (e.g., higher wind speed leads to higher power), but extreme weather samples (D_T) often violate these laws (e.g., high wind speed causing cut-out and zero power). If D_T were excluded from the loss, the model, driven implicitly by D_S , would wrongly predict full power during extreme events. Including D_T in the loss imposes a heavy penalty for such mispredictions, forcing the model to learn the specific mapping of extreme events.
- **Differentiation from MMD:** While MMD brings the “appearance” (features) of normal and extreme weather closer, the Pinball Loss ensures the “content” (labels) is accurate. It forces the regression curve to correctly envelope the rare data points that MMD cannot correct on its own.
- **Asymmetric Weighted Strategy:** We employ a weighted strategy to address the extreme sample imbalance ($n_T \ll n_S$). By setting $C_T \gg C_S$ (e.g., $C_T = 100C_S$), we essentially tell the model that “mispredicting one extreme sample is as grave as mispredicting a hundred normal samples”. This forces the optimizer to prioritize the scarce extreme data, preventing them from being drowned out by the massive conventional data.

3.2.2. Constraints

To handle the non-differentiability of the Pinball Loss, we introduce slack variables ξ_k, ξ_k^* to transform it into linear constraints. For every sample k in the training set $D = D_S \cup D_T$:

$$\begin{cases} y_k - (\mathbf{w}^T \phi(\mathbf{x}_k) + b) \leq \xi_k, & \forall k \in \{1, \dots, n_S + n_T\} \\ (\mathbf{w}^T \phi(\mathbf{x}_k) + b) - y_k \leq \xi_k^*, & \forall k \in \{1, \dots, n_S + n_T\} \\ \xi_k \geq 0, & \xi_k^* \geq 0 \end{cases} \quad (3)$$

In the objective function, the term $L_\tau(y_k, f(\mathbf{x}_k))$ corresponds to $\tau \xi_k + (1 - \tau) \xi_k^*$.

3.3. Model Optimization

In this section, we derive the dual problem of TLQLDMR to enable efficient solving using Quadratic Programming (QP).

3.3.1. Matrix Reformulation of the Primal Problem

To facilitate the derivation, we rewrite the variance term and the MMD term in the objective function [Eq. (2)] using matrix traces or norms. Let $X = [\mathbf{X}_S, \mathbf{X}_T]$ denote the entire training set with $N = n_S + n_T$ samples. For consistency with standard derivations, we add a coefficient of $1/2$ to the variance and MMD terms. The primal objective can be reformulated as:

$$\min_{\mathbf{w}, b, \xi, \xi^*} \frac{1}{2} \|\mathbf{w}\|^2 + \frac{\lambda_1}{2} \text{Var}_S(\mathbf{w}, b) + \frac{\lambda_2}{2} \text{MMD}^2(\mathbf{w}) + \sum_{k=1}^N C_k L_\tau(y_k, f(\mathbf{x}_k)) \quad (4)$$

where C_k takes value C_S if sample $k \in D_S$ and C_T if $k \in D_T$.

We introduce two auxiliary matrices to express the variance and MMD terms in Eq. (4) as quadratic forms of the weights:

1. **Variance Matrix S:** To represent the source domain variance $\sum_{i \in D_S} (y_{S,i} - f(\mathbf{x}_{S,i}) - \bar{u}_S)^2$, we define a centering matrix $\mathbf{H}_S = \mathbf{I}_{n_S} - \frac{1}{n_S} \mathbf{1}_{n_S} \mathbf{1}_{n_S}^T$. The variance term can be written as $\|\mathbf{H}_S(\mathbf{y}_S - \mathbf{f}_S)\|^2$. To extend this to the full dataset, we define an $N \times N$ matrix $\mathbf{S}$:

$$\mathbf{S} = \begin{bmatrix} \mathbf{H}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (5)$$

2. **MMD Matrix $\mathbf{M}$:** The MMD term represents the distance between means. We define the MMD coefficient matrix $\mathbf{M} \in \mathbb{R}^{N \times N}$ as:

$$M_{ij} = \begin{cases} \frac{1}{n_S^2} & i, j \in D_S \\ \frac{1}{n_T^2} & i, j \in D_T \\ \frac{-1}{n_S n_T} & \text{otherwise} \end{cases} \quad (6)$$

3.3.2. Representer Theorem and Kernelization

According to the Representer Theorem [Belkin et al. \(2006\)](#), the optimal weight vector $\mathbf{w}$ lies in the span of the feature mappings of the training samples:

$$\mathbf{w} = \sum_{i=1}^N \beta_i \phi(\mathbf{x}_i) = \Phi \boldsymbol{\beta} \quad (7)$$

where $\Phi = [\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_N)]$ is the feature matrix and $\boldsymbol{\beta} = [\beta_1, \dots, \beta_N]^T$ are the coefficients. Substituting Eq. (7) into the prediction function, we get $f(\mathbf{X}) = \boldsymbol{\beta}^T \mathbf{K} + b \mathbf{1}^T$, where $\mathbf{K} = \Phi^T \Phi$ is the $N \times N$ kernel matrix. Consequently, using Eq. (5) and Eq. (6), the structural risk, variance, and MMD terms can be expressed in terms of $\boldsymbol{\beta}$:

- Structural Risk: $\|\mathbf{w}\|^2 = \boldsymbol{\beta}^T \mathbf{K} \boldsymbol{\beta}$.
- Distribution Variance: The quadratic part with respect to $\boldsymbol{\beta}$ is $\boldsymbol{\beta}^T \mathbf{K} \mathbf{S} \mathbf{K} \boldsymbol{\beta}$. Note that the bias b is eliminated by the centering matrix $\mathbf{H}_S$.
- Domain Adaptation (MMD): $\text{MMD}^2(\mathbf{w}) = \boldsymbol{\beta}^T \mathbf{K} \mathbf{M} \mathbf{K} \boldsymbol{\beta}$.

Combining these, we define the total coefficient matrix $\boldsymbol{\Omega}$:

$$\boldsymbol{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K} \quad (8)$$

By substituting the kernelized terms and Eq. (8) into Eq. (4), the kernelized primal problem becomes (ignoring constant terms w.r.t b in variance):

$$\min_{\boldsymbol{\beta}, b, \xi, \xi^*} \quad \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Omega} \boldsymbol{\beta} - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \boldsymbol{\beta} + \sum_{k=1}^N C_k (\tau \xi_k + (1 - \tau) \xi_k^*) \quad (9)$$

subject to the constraints from Eq. (3):

$$y_k - (\mathbf{K}_k \boldsymbol{\beta} + b) \leq \xi_k, \quad (\mathbf{K}_k \boldsymbol{\beta} + b) - y_k \leq \xi_k^*, \quad \xi_k, \xi_k^* \geq 0 \quad (10)$$

3.3.3. Lagrangian and Dual Problem

We verify the problem by introducing Lagrange multipliers $\alpha_k, \alpha_k^*, \mu_k, \mu_k^* \geq 0$ for Eq. (10). The Lagrangian function $\mathcal{L}$ is constructed as:

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Omega} \boldsymbol{\beta} - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \boldsymbol{\beta} + \sum_{k=1}^N C_k (\tau \xi_k + (1 - \tau) \xi_k^*) \\ & - \sum_{k=1}^N \alpha_k (\xi_k - y_k + \mathbf{K}_k \boldsymbol{\beta} + b) - \sum_{k=1}^N \alpha_k^* (\xi_k^* + y_k - \mathbf{K}_k \boldsymbol{\beta} - b) - \sum (\mu_k \xi_k + \mu_k^* \xi_k^*) \end{aligned} \quad (11)$$

By setting the partial derivatives of Eq. (11) with respect to the primal variables $(b, \xi, \xi^*, \boldsymbol{\beta})$ to zero, we obtain the KKT conditions:

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \implies \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \quad (12)$$

$$\frac{\partial \mathcal{L}}{\partial \xi_k} = 0, \frac{\partial \mathcal{L}}{\partial \xi_k^*} = 0 \implies \alpha_k \in [0, C_k \tau], \quad \alpha_k^* \in [0, C_k(1 - \tau)] \quad (13)$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = 0 \implies \Omega \beta - \lambda_1 KSY - K(\alpha - \alpha^*) = 0 \quad (14)$$

Let $\gamma = \alpha - \alpha^*$. From Eq. (14), the optimal β is derived as:

$$\beta = \Omega^{-1}(K\gamma + \lambda_1 KSY) \quad (15)$$

Substituting Eq. (15) back into Eq. (11), and simplifying using Eq. (12) and Eq. (13), we arrive at the **Dual Problem**, which is a standard Quadratic Programming (QP) problem:

$$\begin{aligned} \min_{\alpha, \alpha^*} \quad & \frac{1}{2}(\alpha - \alpha^*)^T \mathbf{Q}_{dual}(\alpha - \alpha^*) + \mathbf{p}_{dual}^T(\alpha - \alpha^*) \\ \text{s.t.} \quad & \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \\ & 0 \leq \alpha_k \leq C_k \tau, \quad \forall k \\ & 0 \leq \alpha_k^* \leq C_k(1 - \tau), \quad \forall k \end{aligned} \quad (16)$$

where the dual matrix $\mathbf{Q}_{dual}$ and vector $\mathbf{p}_{dual}$ are defined as:

$$\mathbf{Q}_{dual} = K\Omega^{-1}K, \quad \mathbf{p}_{dual} = \lambda_1 K\Omega^{-1}KSY - Y \quad (17)$$

Once the optimal α, α^* are found using a QP solver for Eq. (16), we can recover β using Eq. (15) and construct the final decision function $f(\mathbf{x})$.

3.4. The Algorithm and Complexity Analysis

The complete training process of the proposed TLQLDMR model is summarized in Algorithm 1. The algorithm mainly consists of matrix construction and solving a Quadratic Programming (QP) problem.

3.4.1. Complexity Analysis

The computational complexity of TLQLDMR primarily stems from three parts: matrix multiplication, matrix inversion, and the QP solver.

1. **Matrix Operations:** The construction of $\mathbf{K}$, $\mathbf{S}$, and $\mathbf{M}$ involves $O(N^2d)$ or $O(N^2)$ operations. The calculation of Ω involves matrix multiplications of size $N \times N$, leading to a complexity of $O(N^3)$.
2. **Matrix Inversion:** Inverting the $N \times N$ matrix Ω requires $O(N^3)$ complexity.
3. **QP Solver:** Solving the standard Quadratic Programming problem for the dual variables typically has a complexity ranging from $O(N^{2.3})$ to $O(N^3)$ depending on the solver implementation (e.g., Interior Point Method).

Therefore, the overall time complexity of the proposed algorithm is approximately $O(N^3)$. Given that $N = n_S + n_T$ and $n_T \ll n_S$, the scale is dominated by the source domain sample size. For large-scale datasets, approximation techniques like the Nyström method can be employed to reduce the complexity, which will be considered in future work.

4. Experiments

5. Conclusion

CRedit authorship contribution statement

***: Methodology, Software, Writing - original draft. ***: Visualization, Data curation. ***: Formal analysis, Software. ***: Writing - review & editing. ***: Funding acquisition, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Algorithm 1: TLQLDMR Training Algorithm

Input : Source Data $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, Target Data $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, Quantile τ , Kernel parameters, Regularization parameters λ_1, λ_2 , Pinball Loss weights C_S, C_T .

Output: Optimal dual variables α, α^* and bias b .

// Step 1: Data Preparation

- 1 Combine Source and Target data: $\mathbf{X} = [\mathbf{X}_S, \mathbf{X}_T]$, $\mathbf{Y} = [\mathbf{Y}_S; \mathbf{Y}_T]$, $N = n_S + n_T$;
- 2 Set weights vector $\mathbf{C} \in \mathbb{R}^N$ where $C_k = C_S$ for $k \leq n_S$ and C_T otherwise;

// Step 2: Matrix Construction

- 3 Compute Kernel Matrix $\mathbf{K} \in \mathbb{R}^{N \times N}$ using the specified kernel function;
- 4 Construct Variance Matrix $\mathbf{S}$ using Eq. (5);
- 5 Construct MMD Matrix $\mathbf{M}$ using Eq. (6);

// Step 3: Dual Problem Formulation

- 6 Compute coefficient matrix $\mathbf{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K}$ using Eq. (8);
- 7 Invert $\mathbf{\Omega}$ to fit the dual form;
- 8 Compute Dual Matrix $\mathbf{Q}_{dual} = \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K}$ using Eq. (17);
- 9 Compute Dual Vector $\mathbf{p}_{dual} = \lambda_1 \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{Y}$ using Eq. (17);

// Step 4: Solve QP

- 10 Solve the QP problem in Eq. (16) providing $\mathbf{Q}_{dual}$, $\mathbf{p}_{dual}$, and constraints $0 \leq \alpha_k \leq C_k \tau$, $0 \leq \alpha_k^* \leq C_k(1 - \tau)$;
- 11 **Return** Optimal α, α^* ;

// Step 5: Recover Primal Variables

- 12 Compute $\gamma = \alpha - \alpha^*$;
- 13 Recover β using Eq. (15);
- 14 Compute bias b using complementary slackness conditions (averaged over Support Vectors);

Acknowledgments

This work is supported by the National Key Research and Development Program of China (No. 2021YFB3101500) and the Chongqing Municipal Training Program of Innovation and Entrepreneurship for Undergraduate (No. X202310635204).

References

Belkin, M., Niyogi, P., & Sindhvani, V. (2006). Manifold regularization: A geometric framework for learning from labeled and unlabeled examples. *Journal of machine learning research*, 7.